Project Proposal

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Submitted to-

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OneReserve

Introduction

OneReserve is a Smart Travel and Booking Management System designed to provide an integrated platform for transportation and hotel reservations. The system allows users to select their current location and destination (within Bangladesh or abroad) and suggests suitable travel options such as buses, trains, or airplanes with schedule and route details.After reaching the destination, the system recommends available hotels and provides map-based routes to reach them using local transport options. Users can view travel routes, make reservations, and modify or cancel bookings when needed. The main goal of OneReserve is to simplify travel planning by combining multiple booking services into one efficient and user-friendly database system.

Objectives

- An integrated database system for managing transportation and hotel booking services in one platform.
- To allow users to select their current location and destination (within Bangladesh and internationally).
- To provide suitable travel suggestions (bus, train, airplane) with schedules, routes, and availability details.
- To recommend hotels based on destination, availability, and user preferences.
- To display travel routes and navigation using map-based visualization.
- To enable users to book, modify, and cancel reservations easily.
- To ensure efficient data management, faster booking processes, and a user-friendly travel planning experience.

Project Scope

OneReserve is a comprehensive travel and booking system that allows users to:

- Book bus, train, or airplane tickets with schedules, routes, and availability.
- Reserve hotels at their destination with options to modify or cancel bookings.
- View map-based routes from origin to destination and from transport points to hotels, with local travel suggestions.
- Manage all bookings in one platform efficiently and securely.

Features

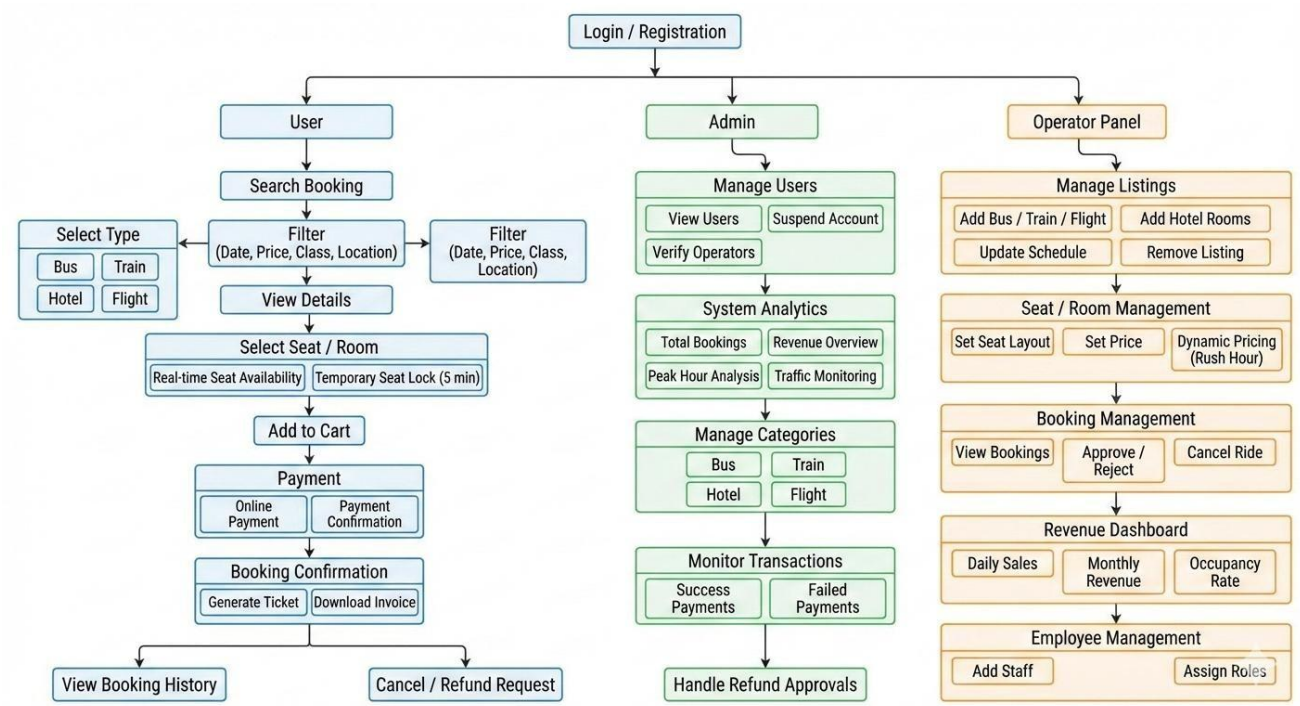
- User Registration & Login – Secure user account creation and authentication.
- Location & Destination Selection – Choose current location and destination (domestic or international).
- Smart Travel Suggestions – Recommend suitable transport (bus, train, airplane) with schedules and routes.
- Transportation Booking – Search, reserve, modify, or cancel tickets.
- Hotel Recommendation & Booking – Suggest and reserve hotels based on destination and availability.
- Route Visualization – Display travel routes and navigation using map-based guidance.
- Local Transport Suggestions – Show bus or car options to reach hotels from arrival points.
- Integrated Booking Management – Manage all travel and hotel reservations in one platform.
- Availability & Price Information – Show real-time availability and booking details.
- User-Friendly Interface – Simple and interactive system for easy travel planning.

System Design

Tools & Technologies:

- Frontend: React.js, HTML, CSS, JavaScript, Bootstrap/Tailwind, Google Maps API for interactive UI and route visualization.
- Backend: Python (Flask or Django) to handle server logic, APIs, and recommendation features.
- Database & Others: MySQL/PostgreSQL for bookings, Git/GitHub for version control, Postman for testing, and Heroku/AWS for deployment.

Diagram:



Conclusion

In conclusion, **OneReserve** aims to provide an efficient and user-friendly travel and booking management system that integrates transportation and hotel services into a single platform. The system simplifies journey planning by offering smart travel suggestions, route visualization, and easy booking management for destinations within Bangladesh and abroad. By ensuring efficient data management and seamless user experience, OneReserve enhances convenience, reduces travel complexity, and improves overall booking efficiency.

Team Member Roles

Member Name and ID	Roles
1.Mahfuz Hossain Antor ID : 24524203158	Lead System Architect & Technical Coordinator
2.Pritoma Paul Lopa ID:24524203170	Backend Lead (Authentication & Admin Control System)
3.Kawser Kafi ID:2452403088	Frontend Integration & System Testing
4.Nahidul Islam Shihab ID:2452403034	Booking Module Support & Documentation Lead